Ahmed Abdelrehim

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Education

Hamilton College — Clinton, NY

Bachelor of Arts in Computer Science, Mathematics; GPA: 3.88 — *Dec. 2024*

- Magna Cum Laude; CS Departmental Honors; ICPC NENA Regional 1st Undergraduate Team (March 2022)
- Study Abroad: The University of Manchester (Jan. 2023 – Jan. 2024)
- Affiliations: Phi Beta Kappa, ColorStack, CodePath, Break Through Tech
- Relevant Coursework: Advanced DS&A, Distributed Systems, AI & ML, Deep Learning, Cognitive Robotics

Experience

Stealth Startup – Second Instructor — San Francisco, CA – Remote

Co-Founder, CTO — *Mar. 2026 – Present*

- Building AI-native LMS for AI-assisted curriculum building, teaching, assesment, grading, and learning insights
- Benchmarking with **10+** models for live teaching, feedback, and grading quality with anti-cheating guards in-place
- Co-leading outreach to professors at top universities to shape product decisions around their teaching practices and invite them to test early prototypes

Meta — New York, NY

Software Engineer — ML Infrastructure — *Feb. 2025 – Present*

- Re-designed the backend orchestrator of the highest-usage workflow for Ads ranking models productionization, reducing processing latency by ∼**88%** and accelerating experiment turnaround for model development
- Developed a generalized workflow to scale a key ML technique to dynamically swap features across dozens of Ads models in production, driving ∼**1.45%** NE gain across relevant prediction tasks
- Introduced a caching layer for a new set of model performance offline metrics, with projected savings of ∼**102** engineering hours annually by reducing repeated computation and data retrieval overhead
- Built an E2E model analysis and evaluation tool for ad-hoc and automated offline metrics statistical analysis, blocking wasteful training jobs and reducing capacity usage by ∼**10%** for adopted models per launch
- Created a platform for dynamic workflows to autonomously generate baseline models for metrics comparison while navigating software and hardware heterogeneity, ultimately speeding up iterations by ∼**2 Wks**
- Co-built a harness to scale E2E agentic iterations on Ads models as part of Meta's Ranking Engineer Agent (REA) program to reduce iteration latency by ∼**50%**, resulting in an engineering blog and Ads all-hands mention

HPE - Juniper Networks — Westford, MA

Software Engineer Intern — *May 2024 – Aug. 2024*

- Rewrote a Python microservice in Go, removing obsolete functionality and reducing code size by **72%**
- Led migration of Address Pool Manager (APM) monitoring to use Prometheus clients instead of terminal logs
- Created custom metric endpoints as parallel processes with instrumentation code and gRPC
- Configured Helm charts to scrape metrics from the K8s cluster running APM and endpoints on known ports
- Built Grafana dashboards to view metrics and Redis-stored values for real-time monitoring

Hamilton College Computer Science Department — Clinton, NY

ML Research Assistant — *May 2022 – Aug. 2022*

- Fine-tuned a Genetic Programming ML model in Clojure, reducing required training cases by ∼**11%**
- Fixed bugs in evaluation functions for ∼**20** benchmark tests, increasing success rate by ∼**10%**
- Automated experiment execution on HPC cluster and results visualization in RStudio with Python

Technical Skills

Programming: Python, C/C++, Golang, PHP, SQL, HTML/CSS, JavaScript, TypeScript, Clojure
ML / Data: PyTorch, scikit-learn, NumPy, Matplotlib, model evaluation, statistical analysis
Backend / Infrastructure: Distributed systems, ML workflow orchestration, GraphQL, REST APIs, gRPC, Docker, Kubernetes (K8s), Redis, MongoDB, AWS
Observability / Tooling: Git, Prometheus, Grafana, PyTest, Linux/Unix, ArgoCD